Binær

3. marts 2022 13:16

$X = N$
Sålænge $X > 0$ gentag:
Næste ciffer = rest ved heltalsdivision $X:2$
X = kvotient ved heltalsdivision $X:2$

Eksempel: $N = 25$:

Heltalsdivision	Kvotient	Rest
25:2	12	1
12:2	6	0
6:2	3	0
3:2	1	1
1:2	0	1

$25 = 11001_2$

13//2 = 6

13 % 2 = 1

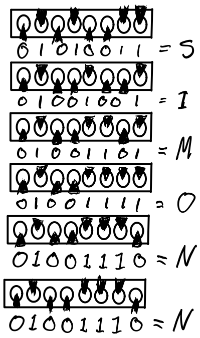
```
while x > 0
    x // 2
print(x % 2)
```

input: 25

11001

Three-bit integers

Bits	Unsigned value	Signed value (Two's complement)
000	0	0
001	1	1
010	2	2
011	3	3
100	4	−4
101	5	−3
110	6	−2
111	7	−1



0111	7	1111	F
0110	6	1110	E
0101	5	1101	D
0100	4	1100	C
0011	3	1011	B
0010	2	1010	A
0001	1	1001	9
0000	0	1000	8

A

gange to bitstrenge sammen

$111 \cdot 101$

$111 = 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 2^2 + 2^1 + 2^0$

$101 = 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = 2^2 + 2^0$

$$111 \cdot 101 = (2^2 + 2^1 + 2^0) \cdot (2^2 + 2^0)$$
$$= (2^2 + 2^1 + 2^0) \cdot 2^2 + (2^2 + 2^1 + 2^0) + 2^0$$
$$= 111 \cdot 100 + 111 \cdot 001 = 11100 + 111 = 100011$$